Day 73 — TNPSC Group 2A Study Material

Part A – General Studies: Current Affairs in Geography of India (4): India's Ramsar Wetland Sites & Tamil Nadu's Lead

READING MATERIAL

Covers India's expanding Ramsar Wetland network and Tamil Nadu's leading position in it — a natural Geography-meets-TN current-affairs topic.

Q: What 2 new Ramsar wetland sites did India add ahead of World Wetlands Day 2026 (2 February), and where are they?

A: Patna Bird Sanctuary (Etah district, Uttar Pradesh — established 1991, 108 hectares, 300+ bird species) and Chhari Dhand Wetland Reserve (Kutch district, Gujarat — a seasonal desert wetland between the Banni grasslands and the Rann of Kutch, home to migratory birds including the Dalmatian Pelican).

Q: How many Ramsar sites did India have immediately after this February 2026 addition, and what expansion does this represent since 2014?

A: 98 Ramsar sites — a 276% expansion since 2014 (when India had only 26).

Q: As of June 2026, what is India's total Ramsar site count, and which state has the most?

A: 100 Ramsar sites nationally (covering ~13,876 sq. km); Tamil Nadu has the most of any state, with 20 sites covering 1,008.81 sq. km.

Q: Name 3 of Tamil Nadu's well-known Ramsar sites.

A: Gulf of Mannar Marine Biosphere Reserve, Vedanthangal Bird Sanctuary, and Pichavaram Mangrove (among TN's 20 total).

MOCK TEST (WITH ANSWERS)

1. Which 2 wetlands were added to India's Ramsar list ahead of World Wetlands Day 2026?

A. Vedanthangal & Pichavaram

B. Patna Bird Sanctuary & Chhari Dhand Wetland Reserve ✓

C. Sundarbans & Chilika Lake

D. Gulf of Mannar & Loktak Lake

Explanation: Patna Bird Sanctuary (UP) & Chhari Dhand Wetland Reserve (Gujarat). [hard, web-research]

2. Chhari Dhand Wetland Reserve is located in which state?

A. Rajasthan

B. Gujarat ✓

C. Madhya Pradesh

D. Maharashtra

Explanation: Gujarat (Kutch district). [medium, web-research]

3. After the February 2026 addition, India's Ramsar site count reached 98 — representing what expansion since 2014?

A. 100%

B. 150%

C. 276% ✓

D. 320%

Explanation: 276% expansion since 2014. [hard, web-research]

4. As of June 2026, which state has the highest number of Ramsar sites in India?

A. Uttar Pradesh

B. Odisha

C. Tamil Nadu ✓

D. Gujarat

Explanation: Tamil Nadu (20 sites). [medium, web-research]

5. As of June 2026, Tamil Nadu has how many Ramsar sites?

A. 12

B. 16

C. 20 ✓

D. 24

Explanation: 20. [medium, web-research]

6. Which of these is NOT one of Tamil Nadu's Ramsar sites?

A. Gulf of Mannar Marine Biosphere Reserve

B. Vedanthangal Bird Sanctuary

C. Pichavaram Mangrove

D. Chhari Dhand Wetland Reserve ✓

Explanation: Chhari Dhand is in Gujarat, not Tamil Nadu. [easy, web-research]

Part B – Aptitude & Mental Ability: Volume

READING MATERIAL

Volume is the last of 10 named Aptitude sub-skills. Real exam evidence shows two dominant shapes: 'ratio of dimensions → ratio of volumes' (scaling), and 'melt/recast one solid into another shape' (volume is conserved, surface area is not) — both are core techniques worth knowing cold.

Q: Method — ratio of volumes from a ratio of radii (same shape): If the radius of one sphere is half that of another, what is the ratio of their volumes?

A: Volume scales with the CUBE of linear dimensions ($V=4/3\pi r^3$), not directly with the radius. So if $r_1:r_2=1:2$, then $V_1:V_2 = 1^3:2^3 = 1:8$.

Q: Method — melting and recasting (volume is conserved, NOT surface area): A sphere of radius 9cm is melted and recast into a cone of the same radius (9cm). Find the cone's height.

A: Set sphere volume = cone volume: $(4/3)\pi(9)^3 = (1/3)\pi(9)^2h \rightarrow (4/3)(729) = (1/3)(81)h \rightarrow 972 = 27h \rightarrow h = 36$ cm. The general shortcut: for melting sphere→cone of the SAME radius, height = 4×radius always (since the $(4/3)r^3$ vs $(1/3)r^2h$ relationship reduces to $h=4r$ when radii match).

Q: Real exam (already seen in an earlier topic, re-noted here): A sphere of diameter 8cm is melted and drawn into a wire of diameter 3mm. Find the wire's length.

A: Volume conserved: $(4/3)\pi(4)^3 = \pi(0.15)^2\times h \rightarrow h \approx 3792.6$ cm ≈ 37.9 m. (Same melt/recast principle as r02, just sphere→cylinder/wire instead of sphere→cone.)

Q: Method — volume of a cylindrical tank from its capacity: A cylindrical tank has capacity 1,848 m^3 and base diameter 14m. Find its depth.

A: $V=\pi r^2h \rightarrow 1848 = (22/7)\times7^2\times h = 154h \rightarrow h = 12$ m.

Q: Method — dug-out soil redistributed into a different shape (volume conserved): A cylindrical well of depth 20m, diameter 14m is dug; the soil is spread evenly to form a cuboid platform of base 20m×14m. Find the platform's height.

A: Well volume = $(22/7)\times7^2\times20 = 3{,}080$ m^3. Platform volume = $20\times14\times h = 280h$. Set equal: $280h=3080 \rightarrow h=11$ m.

Q: Formula check: volume of a cube with side 5cm, and the relationship between a cube's volume and surface area.

A: Volume = $\text{side}^3 = 5^3 = 125$ cm^3. If a cube's volume is 1000 cm^3, its side = $\sqrt[3]{1000}$ = 10cm, so its surface area = $6\times\text{side}^2 = 6\times100 = 600$ cm^2 (cube surface area = $6\times\text{side}^2$, from its 6 square faces).

Q: Method — ratio of volumes for cone, sphere, cylinder of equal radius AND equal height: What is that ratio?

A: Cone : Sphere : Cylinder = 1 : 4 : 3, when all three share the same radius r and the same height h (for the sphere, 'height' = its diameter = 2r in this comparison). This is a frequently tested ratio worth memorizing directly.

MOCK TEST (WITH ANSWERS)

1. If the radius of sphere A is 3 times the radius of sphere B, what is the ratio of their volumes (A:B)?

A. 3:1

B. 9:1

C. 27:1 ✓

D. 6:1

Explanation: Volume ratio = $(\text{radius ratio})^3 = 3^3:1^3 = 27:1$. [easy, authored]

2. A solid sphere of radius 6cm is melted and recast into a cone of the same radius. Find the cone's height.

A. 18 cm

B. 24 cm ✓

C. 12 cm

D. 6 cm

Explanation: Using the shortcut h=4×radius for sphere→same-radius-cone: h=4×6=24cm. [medium, authored]

3. A cylindrical tank has capacity 4,620 m^3 and base diameter 14m. Find its depth.

A. 20 m

B. 25 m

C. 30 m ✓

D. 15 m

Explanation: $V=\pi r^2 h \rightarrow 4620 = (22/7)\times 49\times h = 154h \rightarrow h=30m$. [medium, authored]

4. Find the volume of a cube with side 8cm.

A. 512 cm³ ✓

B. 256 cm³

C. 64 cm³

D. 640 cm³

Explanation: $8^3 = 512$ cm³. [easy, authored]

5. If a cube's volume is 2,197 cm³, find its surface area.

A. 507 cm²

B. 169 cm²

C. 1014 cm² ✓

D. 338 cm²

Explanation: Side = $\sqrt[3]{2197}$ = 13cm. Surface area = $6\times 13^2 = 6\times 169 = 1,014$ cm². [medium, authored]

6. A cone, a sphere, and a cylinder share the same radius and same height. What is the ratio of their volumes (cone:sphere:cylinder)?

A. 1:3:4

B. 1:4:3 ✓

C. 3:4:1

D. 4:3:1

Explanation: Cone:Sphere:Cylinder = 1:4:3 for equal radius and height. [medium, web-research]

Part C – General English: Translation: Tamil-English Sentence, Word, and Tense/Voice Translation

READING MATERIAL

Tests both Tamil-to-English and English-to-Tamil sentence translation, with TENSE and VOICE precision as the main skill being checked — plus basic translation terminology (literal vs. literary vs. machine translation).

Q: Real exam: translate the Tamil sentence meaning 'By this time tomorrow, I will have completed this work' (future-perfect-tense sense) into English. Which of these is correct: (A) I will have completed this work by tomorrow (B) I would have finished this work by tomorrow (C) I have completed this work by this time tomorrow (D) I would have completed this work by this time tomorrow?

A: (D) — 'I would have completed this work by this time tomorrow.' The TNPSC-marked answer uses 'would have' (not 'will have') to match the original's modal nuance, AND keeps the exact 'by this time tomorrow' phrase rather than shortening it to just 'tomorrow' — both details matter for picking the precisely correct option over close-sounding distractors.

Q: Real exam: 'Sentence: They were being followed when they left the building.' Which Tamil sentence best conveys this English sentence's meaning?

A: Only option (b) was marked correct among 4 close Tamil paraphrases. This tests recognising the PASSIVE VOICE + PAST CONTINUOUS combination ('were being followed') precisely — a common trap is a Tamil option that conveys active voice or a different tense, sounding similar but changing who is doing the action.

Q: Real exam: translate the Tamil sentence meaning 'The birds are flying' (present continuous) into English. Which is correct: (A) The birds are flying (B) Birds fly (C) The birds were flying (D) The birds will fly?

A: (A) 'The birds are flying' — present continuous in the Tamil original must map to present continuous in English, not simple present (B, a general/habitual-truth statement), past continuous (C), or future (D).

Q: Real exam: what is word-for-word translation called?

A: Literal translation (as distinct from 'literary translation' — which preserves style/tone/artistic effect rather than exact word order — and 'machine translation' or 'administrative translation').

Q: Real exam: pick the correct English word/phrase for the given Tamil phrase. (Word/phrase-level translation, not full-sentence.)

A: Correct answer marked: 'Don't induce me' — over close-sounding distractors like 'Don't indulge me' and 'Don't talk to me'. Word/phrase translation questions often have several plausible-sounding English options; precision in the Tamil phrase's exact sense (not just a related English idiom) determines the correct choice.

MOCK TEST (WITH ANSWERS)

1. Translate (future-perfect sense, 'by this time tomorrow, I will have completed this work'):

A. I will have completed this work by tomorrow.

B. I would have finished this work by tomorrow.

C. I have completed this work by this time tomorrow.

D. I would have completed this work by this time tomorrow. ✓

Explanation: I would have completed this work by this time tomorrow. [hard, question-bank-2025]

2. 'They were being followed when they left the building.' This English sentence uses which combination of voice and tense?

A. Active voice, simple past

B. Passive voice, past continuous ✓

C. Active voice, past continuous

D. Passive voice, simple past

Explanation: Passive voice, past continuous ('were being followed'). [hard, question-bank-2025]

3. Translate (present continuous sense, 'the birds are flying'):

A. The birds are flying ✓

B. Birds fly

C. The birds were flying

D. The birds will fly

Explanation: The birds are flying. [medium, question-bank-2025]

4. What is word-for-word translation called?

A. Literary translation

B. Literal translation ✓

C. Administrative translation

D. Machine translation

Explanation: Literal translation. [easy, question-bank-2025]

5. A translation style that preserves style, tone, and artistic effect rather than exact word order is called:

A. Literal translation

B. Literary translation ✓

C. Machine translation

D. Administrative translation

Explanation: Literary translation. [medium, authored]

6. Pick the correct English meaning for the given Tamil phrase (real exam item testing precise word/phrase-level translation):

A. Don't induce me ✓

B. Don't indulge me

C. Don't talk to me

D. Don't make me do what I don't want to

Explanation: Don't induce me. [medium, question-bank-2025]